

Initial Mechanical Design of a Simple Robot Dog

Prepared by:

Aisha ALhuraysi

Summer Training 2026

Smart Methods

Body Shape and Structure

The robot has a simple rectangular body that acts as the main frame. This shape was chosen because it is easy to design, lightweight, and provides enough space for future electronic components such as the battery and controller.

The body supports four legs that are attached at the four corners to improve balance and weight distribution.

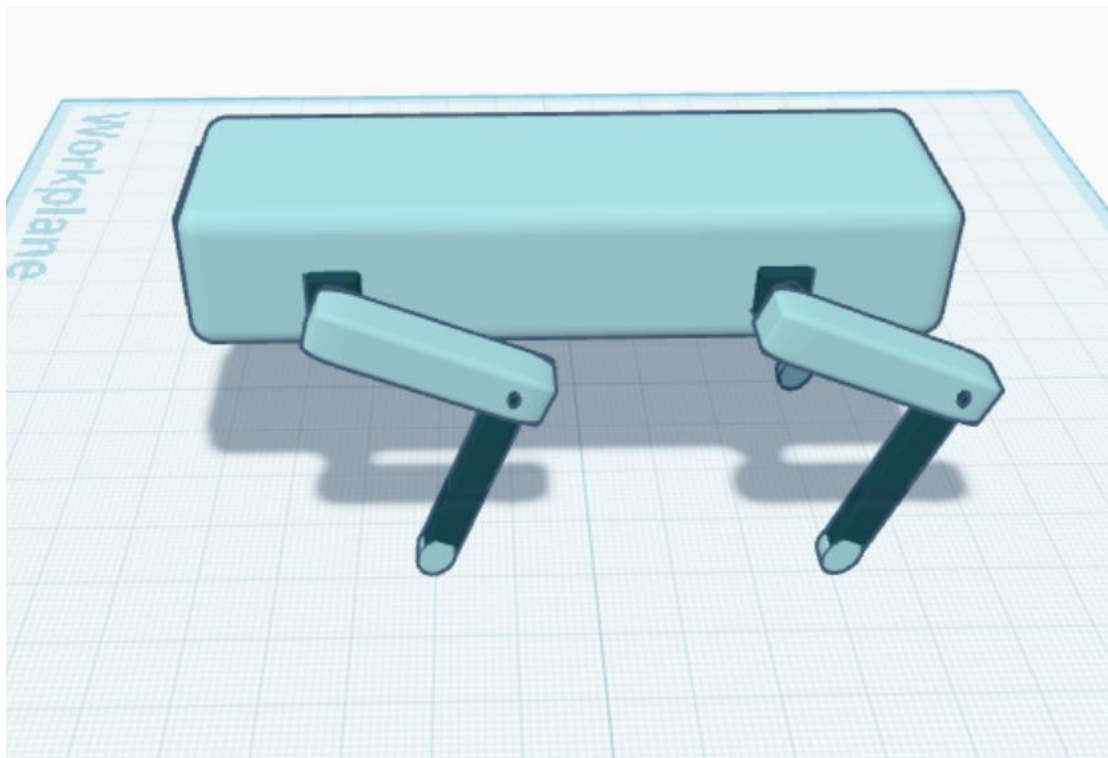


Figure 1: *Perspective view of the complete robot design.*

Leg Design

The robot has four identical legs. Each leg consists of:

- Upper leg
- Lower leg

This simple design allows the robot to perform basic walking movements while keeping the structure lightweight.

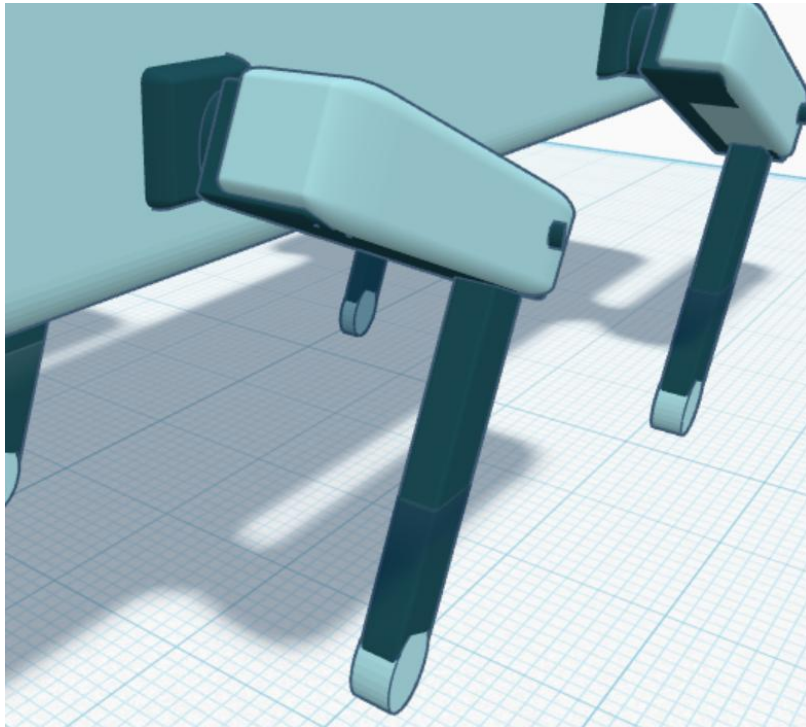


Figure 2: *Close-up view of one robot leg.*

Joints and Degrees of Freedom (DOF)

Each leg has two joints:

- Hip Joint
- Knee Joint

Therefore:

- 2 DOF for each leg
- Total DOF = 8

This number of joints provides enough movement for a basic walking robot.

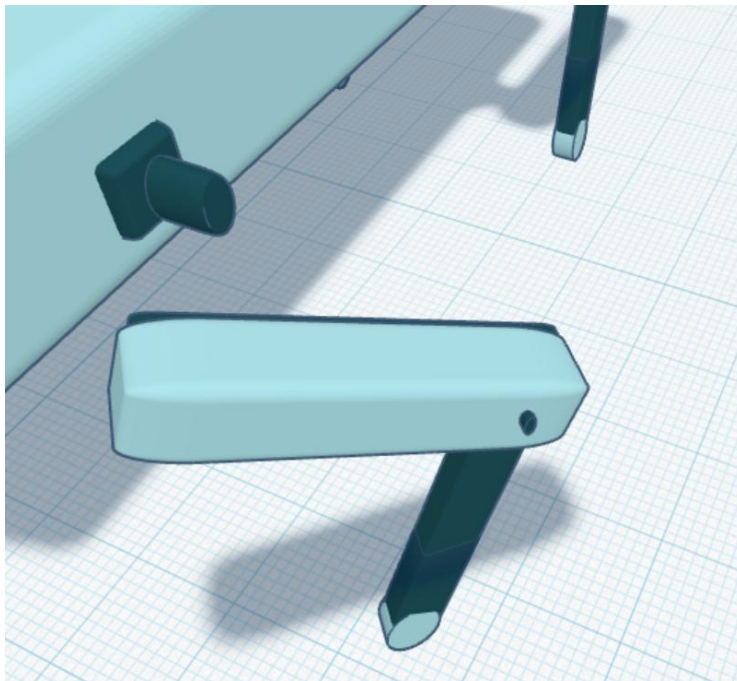


Figure 3: Side view showing the joints.

Motor Selection

Servo motors are selected because they provide accurate position control and are commonly used in small robotic systems.

Recommended motor:

MG996R Servo Motor

Reasons for selection:

- Good torque
- Easy to control

- Affordable
- Suitable for educational robot projects

Each joint requires one servo motor.

Total motors required:

8 Servo Motors

Initial Torque Calculation

The torque is calculated using the following equation:

$$\tau = F \times r$$

Where:

- Mass = 1 kg
- Distance = 0.3 m

Force:

$$F = m \times g$$

$$F = 1 \times 9.81 = 9.81N$$

Torque:

$$\tau = 9.81 \times 0.3 = 2.94 \text{ N} \cdot \text{m}$$

The selected servo motor should provide at least **2.94 N·m** of torque.

Stability and Center of Gravity

The center of gravity is located near the middle of the robot body. Keeping the battery and other components close to the center improves stability and reduces the possibility of tipping while walking.

The four-leg arrangement also helps distribute the robot's weight evenly.

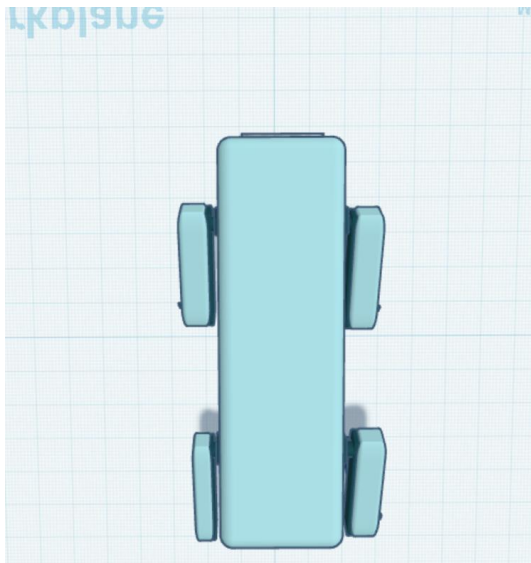


Figure 4: *Top view of the robot.*

Proposed Walking Method

The robot uses a simple walking gait.

Walking sequence:

1. Front Left
2. Rear Right
3. Front Right
4. Rear Left

This sequence keeps three legs on the ground most of the time, improving stability during movement.

Expected Mechanical Problems

Some possible mechanical problems include:

- Limited motor torque
- Joint friction
- Robot imbalance
- Leg bending under heavy loads
- Limited walking speed